



TAS LAB

Trustworthy AI and
Autonomous Systems Lab

THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大学

Department of
Aeronautical and Aviation Engineering
航空及民航工程學系

Remarkable Team Milestones in 2025

- 4 PhD graduates; PolyU–LinXAI Joint Laboratory established
- LAE regulatory sandbox project approved; first sandbox test flights
- Hosted the inaugural RCUAS research workshop

2025 TASLAB Annals

Webpage: <https://polyu-taslab.github.io>

Trustworthy AI and Autonomous Systems Lab, PolyU

19

journal papers

7

conference papers

46

members & visitors

1,041

Google Scholar citations

6

awards & recognitions

70

news stories on our website

Faculty, Postdocs & Staff (PI, Postdoc, RA, Visiting)



Dr Weisong Wen
Asst Prof, AAE



Dr Feng Huang
Postdoc



Dr Runqiu Yang
Postdoc



Dr Shiyu Bai
Postdoc



Dr Yingying Wang
Postdoc



Dr Lu Bai
Postdoc



Dr Jianhao Jiao
Postdoc



Mr Pei Ma
RA



Ms Akida Tursun
Staff, PolyU-Wuxi



Mr Yun Wang
RA, PolyU-Wuxi



Mr Haobo Wu
RA, PolyU-Wuxi



Mr Heng Li
RA



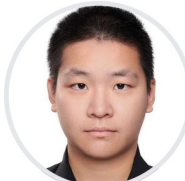
Mr Yuxiang Shi
Visiting, Tongji



Mr Zihao Wang
Visiting, WHU

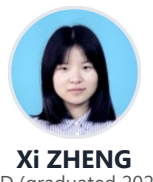


Mr Fengchi Zhu
Visiting, HEU



Mr Guangyan Guo
Visiting, HEU

PhD/MPhil Student Members (and Undergraduates)



Xi ZHENG
PhD (graduated 2025)



Xikun LIU
PhD (graduated 2025)



Yihan ZHONG
PhD (graduated 2025)



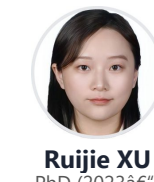
Runzhi HU
PhD (graduated 2025)



Peiwen YANG
PhD (2023&€")



Yifeng CAO
PhD (2023&€")



Ruijie XU
PhD (2023&€")



Yuanyuan ZHANG
PhD (2023&€")



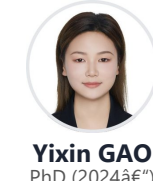
Naigui XIAO
PhD (2024&€")



Bingxian LI
PhD (2024&€")



Ziqi ZHANG
PhD (2024&€")



Yixin GAO
PhD (2024&€")



Xiangru WANG
PhD (2025&€")



Hongchang CHEN
PhD (2025&€")



Jiahao HU
PhD (2025&€")



Xin WANG
PhD (2025&€")



Jiagui ZHONG
PhD (2025&€")



Yu FENG
PhD (2025&€")



Jiaqi ZHAO
MPhil (2024&€")



Yingming CHEN
MPhil (2024&€")



Shaoting QIU
MPhil (2024&€")



Yuling CHENG
MPhil (2024&€")



Qijun QIN
MPhil (2025&€")



Mokui YANG
MPhil (2025&€")



Qi ZHANG
MPhil (2025&€")



Yiyu TANG
MPhil (2025&€")



Qian YANG
MPhil (2025&€")



Junzhe WANG
MPhil (2025&€")



Xinrui ZHANG
MPhil (2025&€")



Zhongqi WANG
UG (2025&€")

2025 in Review

2025 was the lab's busiest year so far, with 70 news stories on our website. Four PhD students completed their studies: Xi Zheng, Xikun Liu, Yihan Zhong and Runzhi Hu. PolyU and LinXAI set up a joint lab for space robotics and embodied AI, our drones entered Hong Kong's low-altitude regulatory sandbox, and PolyU hosted the inaugural RCUAS workshop. The team briefed the Secretary for Transport and Logistics, the Chief Executive's Policy Unit and LegCo members, and published 19 journal and 7 conference papers.

Selected papers

- X. Liu, W. Wen, L. Zhang, L.-T. Hsu. 3D LiDAR aided GNSS NLOS correction by direction-of-arrival estimation using Doppler. IEEE T-ITS
- R. Hu, P. Xu, Y. Zhong, W. Wen. pyrtklib: An open-source package for tightly coupled deep learning and GNSS integration. IEEE T-ITS
- Y. Zhang, Y. Wang, P. Yan, W. Wen. Learning safe, optimal and real-time flight interaction with deep confidence-enhanced reachability. IEEE T-ITS
- P. Yang, W. Wen, S. Bai, L.-T. Hsu. Tightly joined positioning and control model for UAVs based on factor graph optimization. IEEE TVT
- F. Huang, W. Wen, G. Zhang, D. Su, Y. Huang. Continuous error map-aided adaptive multisensor integration for connected AVs. IEEE TIM

Research & Innovation Awards

World's Top 2% Scientists

Stanford University / Elsevier · third year



Silver Medal, 51st Geneva Inventions

International Exhibition of Inventions of Geneva

Research Grant Achievement Award

PolyU Faculty of Engineering

Innovation Award, Angel Fund (TSSSU-O)

Lab spin-off CeresRobotics.ai

Second Prize & Best Presentation

PolyU International Future Challenge (Shanghai)



Best Presentation, GSCS & ICUI 2025

Ruijie Xu (PhD)

International Journal & Conference Publications

IEEE T-ITS IEEE Transactions on Intelligent Transportation (4)	IEEE TIM IEEE Transactions on Instrumentation (3)	IEEE T-IV IEEE Transactions on Intelligent Vehicles (3)	IEEE Sensors IEEE Sensors Journal (2)	IEEE TAES IEEE Transactions on Aerospace and Electronic Systems (2)	IEEE IoT-J IEEE Internet of Things Journal (1)	IEEE TCE IEEE Transactions on Consumer Electronics (1)	IEEE TMC IEEE Transactions on Mobile Computing (1)
IEEE TVT IEEE Transactions on Vehicular Technology (1)	ION NAVI NAVIGATION, Journal of the ION (1)	CONF. ITSC Conference papers (3)	CONF. PLANS Conference papers (2)	CONF. CCC Conference paper (1)	CONF. IROS Conference paper (1)		

Talks, Visits & Service

Feb ASTRI Tech Summit, LAE showcase · Hong Kong

Feb Talk to HKGCC members and lab visit · PolyU

Mar Demo for the Secretary for Transport & Logistics · PolyU

Apr AI4X conference · Hong Kong

Apr Cleaning UAV at InnoEX · Hong Kong

May Briefing, Chief Executive's Policy Unit · Hong Kong

May IEEE/ION PLANS 2025 · Salt Lake City

Jun Nanchang Health Expo · Nanchang

Jun HKGCC panel on the LAE · Hong Kong

Jul Huawei CTO: AI-PDR · Hong Kong

Jul LegCo briefing · Hong Kong

Aug BUAA: safety-assured UAVs · Beijing

Aug CIC briefing · Hong Kong

Aug GSCS & ICUI 2025 · Hong Kong

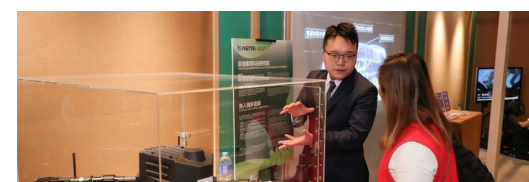
Oct IEEE IROS 2025 · Hangzhou

Oct University of Macau · Macau

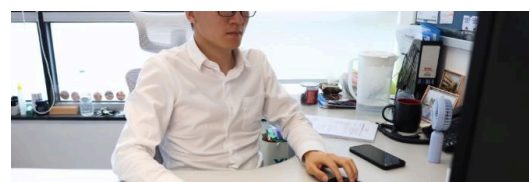
Nov Workshop chair, ITSC 2025 · Gold Coast

Nov 28th Beijing–HK Economic Cooperation Symposium · Hong Kong

Media Publicity



RTHK at HK SciFest: medical delivery drone (Apr)



UBeat: drone-based curtain-wall cleaning (Nov)



OHKF "Taking Flight" LAE report (Sep)

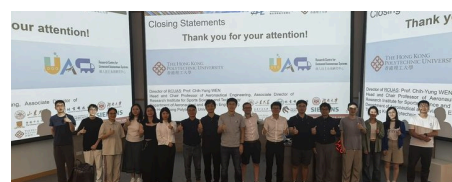


Cleaning-drone demo for the Secretary for T&L (Mar)

Group Gathering / Impact Story



PolyU–LinXAI Joint Laboratory (Mar)



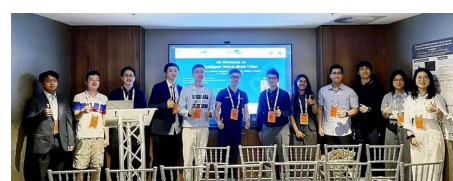
Inaugural RCUAS workshop (Jun)



Sandbox test flight on campus (Sep)



Meituan MarsTalk at PolyU (Sep)



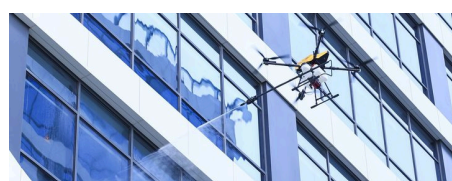
4th IV Meets Urban workshop, ITSC for T&L (Mar)



Dr. Xi Zheng's PhD defence (Nov)



HEU Vice President visits TAS Lab (Nov)



Drone building cleaning, Shanghai (Dec)